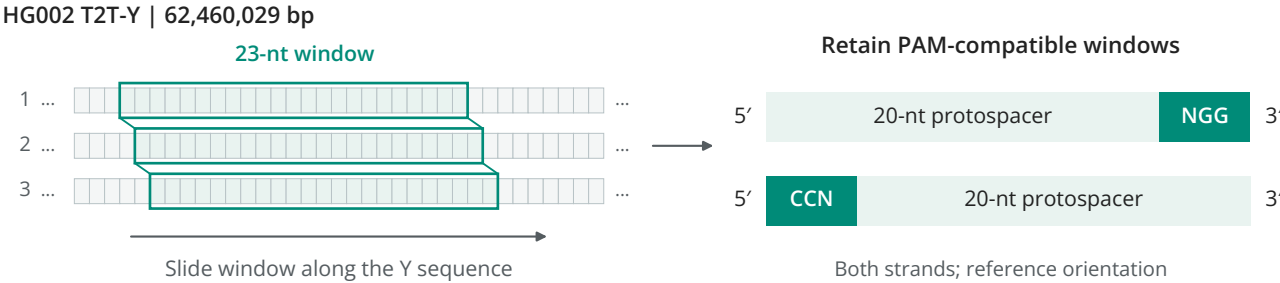
a Scan T2T-Y for PAM-compatible guide sequences



b Candidate screening and the guide shortlist (20-nt)

Assess Y-target abundance and genome-wide specificity

Candidate table
14,723
distinct guides
Both orientations

→

Exact-match filters
6,901
eligible guides
≥10 exact Y matches
0 exact non-Y matches

⇢

Guide shortlist
215
shortlisted guides
No recorded non-Y sites
with 0, 1 or 2 mismatches

Shortlisted guides ranked by decreasing exact Y-match count

c Five guides selected for initial testing

Yq repeats	Centromere		Yp	
Q53	CN	CP	P47	P51
TCGCCAGGAATTGATGTGAA	ATGATAGGTTGAACTCCCGG	ACTATCTTCGCATGAGAACC	TTCGCGCGCAGTCCCTTAGG	ATTAAGGCCCACCTGAGTAA
532	53	54	47	51

Exact Y matches (both orientations); guide sequences shown 5′ to 3′, without PAM